

Junying Fang (方俊颖)

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CURRENT POSITION

Assistant Research Scientist, Guangzhou Institute of Tropical and Marine Meteorology, China Meteorological Administration

July 2023 - Present

EDUCATION

Ph.D. in Meteorology, Sun Yat-sen University, 2023

GPA: 3.86/4 (Rank: 1/24)

Thesis: The diurnal offshore propagation of global coastal rainfall

Advisor: Prof. Yu Du

B.S. in Atmospheric Sciences, Nanjing University of Information Science & Technology, 2017

RESEARCH INTERESTS

- Meteorology
- AI for science
- Autonomous discovery

PUBLICATIONS

1. **Junying Fang**, Yu Du. A global survey of diurnal offshore propagation of rainfall. *Nature Communications*. 2022. <https://doi.org/10.1038/s41467-022-34842-0>
2. Kuai Dai, Xutao Li, **Junying Fang**, et al. Four-hour thunderstorm nowcasting using a deep diffusion model for satellite data. *PNAS*. 2025. <https://doi.org/10.1073/pnas.2517520122>
3. Hongpei Yang, Yu Du, Zijian Chen, **Junying Fang**. Could Developing Frontal Rainfall Influence Warm-sector Rainfall? *Geophysical Research Letters*. 2024. <https://doi.org/10.1029/2024GL110430>
4. **Junying Fang**, Yu Du. Global coastal terrains push rainfall offshore. In preparation.
5. **Junying Fang**, Yu Du. The contribution of MCS on diurnal offshore propagation of rainfall. In preparation.

RESEARCH GRANTS

1. 2025-2027 **National Natural Science Foundation of China**, PI. The Influence of Global Coastal Topography on the Diurnal Offshore Propagation of Rainfall and Its Mechanisms, ¥300,000 RMB
2. 2023-2026 **Guangdong Provincial Fund for Basic and Applied Basic Research**, PI. The Characteristics and Mechanisms of the Diurnal Variation and Offshore Propagation of Mesoscale Convective Systems along the South China Coast, ¥100,000 RMB
3. 2024-2025 **Institute of Tropical and Marine Meteorology, China Meteorological Administration**, PI. Research on Coastal Heavy Rainfall Propagation in the Guangdong-Hong Kong-Macao Greater Bay Area Based on Multi-Source Observations and AI, ¥100,000 RMB
4. 2024-2025 **Guangdong Meteorological Administration**, PI. Research and Forecasting on Multi-Scale Propagation Characteristics of Heavy Rainfall in the Guangdong-Hong Kong-Macao Greater Bay Area, ¥10,000 RMB

5. 2025-2028 **National Natural Science Foundation of China**, Participant. The Effects of Complex Terrain on the Activity Characteristics of Convectively Generated Gravity Waves and Their Roles in the Occurrence and Development of Convection, ¥480,000 RMB
6. 2021-2024 **National Natural Science Foundation of China**, Participant. Propagation Characteristics of Global Coastal Precipitation Diurnal Variation and Its Connection with Gravity Waves and Density Currents, ¥580,000 RMB
7. 2020-2023 **National Natural Science Foundation of China**, Participant. Climatological Characteristics of Mesoscale Convective Systems in China and Their Contributions to Precipitation and Severe Weather Variability with Associated Mechanisms, ¥630,000 RMB
8. 2020-2023 **Guangzhou Basic and Applied Research Project**, Participant. Propagation Characteristics of Global Coastal Precipitation Diurnal Variation and Its Relationship with Gravity Waves, ¥200,000 RMB

PRESENTATIONS

- Jan. 2024, Generative AI Advances Science Research Workshop, Guangzhou, China (oral)
- Aug. 2023, the Meteorological Journal Development Seminar, Zhuhai, China (invited oral)
- Aug. 2023, the Young Meteorological Scientists Conference, Zhuhai, China (oral)
- Aug. 2023, the 5th Chinese Mesoscale Meteorology Forum, Yinchuan, China (oral)
- Jan. 2023, the Workshop of South China Monsoon Heavy Rainfall, Online (invited oral)
- Dec. 2022, the 58th Atmospheric Sciences Forum of Peking University, Online (invited oral)
- Aug. 2022, the 4th Chinese Mesoscale Meteorology Forum, Hangzhou, China (poster)
- May. 2021, the Heavy Rainfall Donghu Forum, Wuhan, China (oral)
- Apr. 2021, the 14th Conference on Mesoscale Convective Systems and High-Impact Weather in East Asia (ICMCS), Nanjing, China (poster)
- Jan. 2021, the 101st Annual Meeting of American Meteorological Society, Online (oral)
- Jan. 2021, the 3rd Chinese Mesoscale Meteorology Forum, Nanjing, China (poster)
- Dec. 2020, the 4th Workshop of Severe Weather International Consortium (SWIC), Zhuhai, China (poster)
- Nov. 2020, the Young Scientist Forum, Fuzhou, China (poster)
- Nov. 2019, the 2nd Chinese Mesoscale Meteorology Forum, Xiamen, China (poster)
- Mar. 2019, the 13th Conference on Mesoscale Convective Systems and High-Impact Weather in East Asia (ICMCS), Okinawa, Japan (poster)
- Dec. 2018, the Atmospheric Science Graduate Academic Forum, Guangzhou, China (oral)
- Oct. 2018, the 35th Annual Meeting of Chinese Meteorological Society, Hefei, China (oral)

AWARDS

- Oral presentation award at the 101st Annual Meeting of American Meteorological Society, 2021
- First-class PhD scholarship of Sun Yat-sen University, 2019
- Oral presentation award at the 35th Annual Meeting of Chinese Meteorological Society, 2018
- Oral presentation award at the Atmospheric Science Graduate Academic Forum, 2018

TEACHING

- Aug. 2020 - Jan. 2021: Teaching assistant for **Synoptic Meteorology**, Sun Yat-sen University
- Feb. 2021 - Jul. 2021: Teaching assistant for **Climate Change**, Sun Yat-sen University
- Aug. 2021 - Jan. 2022: Teaching assistant for **Climate Impacts, Adaptation and Mitigation**, Sun Yat-sen University

SERVICE

- Journal Review: Geophysical Research Letters, Journal of the Atmospheric Sciences, Journal of Geophysical Research: Atmospheres, International Journal of Climatology, Earth and Space Science, Journal of Meteorological Research, Ocean-Land-Atmosphere Research
- Journal Editorial Intern: The Innovation